CHAPTER 5 FIRE SERVICE FEATURES

General Comments

The requirements of this chapter apply to all occupancies and pertain to access roads, access to building openings and roofs, premises identification, key boxes, hazards to firefighters, fire protection water supplies, fire command centers, fire department access to equipment and in-building emergency responder communications enhancement system (ERCES) coverage.

Purpose

This chapter contains the requirements for fire service access to the property that is to be protected, including access roads, security devices and access through openings in the building.

The chapter also addresses firefighter hazards, the requirements for a fire department command center and firefighter access to equipment, such as fire suppression equipment, air-handling equipment, emergency power equipment and access to the roof. In addition, this chapter addresses the fire protection water supply.

Finally, this chapter addresses in-building emergency responder communications enhancement system coverage so that emergency personnel can communicate during a fire or emergency at the scene of the fire.

SECTION 501—GENERAL

501.1 Scope. Fire service features for buildings, structures and premises shall comply with this chapter.

C This chapter contains requirements that will enable the fire service to respond to an emergency on the premises of a building or structure.

501.2 Permits. A permit shall be required as set forth in Sections 105.5 and 105.6.

C Permits must be obtained from the fire code official. Permit fees, if any, must be paid prior to the issuance of the permit. There are two types of permits: operational and construction. The operational permits required by this section are for the use or operation of fire protection valves and fire hydrants (see Section 105.5.17) or the use or removal from service of a private fire hydrant (see Section 105.5.43). The construction permit (see Section 105.6.19) allows the applicant to install or modify private fire hydrants. See Section 105 for additional information on permits.

501.3 Construction documents. *Construction documents* for proposed fire apparatus access, location of *fire lanes*, security gates across fire apparatus access roads and *construction documents* and hydraulic calculations for fire hydrant systems shall be submitted to the fire department for review and approval prior to construction.

C The integrity of the design of fire apparatus access roads and private fire hydrant systems is critical to successful fire department operations in protecting lives and property. Construction documents must be drawn to scale and clearly show the details of the indicated features in order for the fire department to properly evaluate fire service features and issue approvals. While this section is primarily for new installations, if a security gate was not on the original construction plans for fire apparatus access, this section reinforces that a security gate installed at a later time requires a construction plan to be submitted and approved prior to construction.

501.3.1 Site safety plan. The *owner* or *owner's* authorized agent shall be responsible for the development, implementation and maintenance of an *approved* written *site safety plan* in accordance with Section 3303.

C This section places the responsibility for a site safety plan, as required by Section 3303, on the owner. This section is appropriately located under construction documents. The site safety plan is required to document items such as procedures for reporting emergencies; fire department access routes; and locations of fire protection equipment, including fire department connections and fire hydrants.

501.4 Timing of installation. Where fire apparatus access roads or a water supply for fire protection are required to be installed, such protection shall be installed and made serviceable prior to and during the time of construction except where *approved* alternative methods of protection are provided. Temporary street signs shall be installed at each street intersection where construction of new roadways allows passage by vehicles in accordance with Section 505.2.

C Buildings under construction are quite vulnerable to fire and other types of construction incidents, such as injuries from falling objects. Access roads and water for fire protection are essential for firefighting purposes. Temporary street signs are also valuable to emergency responders because the streets in new developments will most likely not be familiar to them or be on their maps.

Marked access roads and an emergency water supply should be in place before any large amount of combustible building materials is placed on site and before any construction is initiated.

SECTION 502—DEFINITIONS

502.1 Definitions. The following terms are defined in Chapter 2:

AGENCY.

FIRE APPARATUS ACCESS ROAD.

FIRE COMMAND CENTER.

FIRE DEPARTMENT MASTER KEY.

FIRE LANE.

KEY BOX.

TRAFFIC CALMING DEVICES.

C Definitions of terms can help in the understanding and application of code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2, or in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (also see the commentaries to Sections 201.1 through 201.4).

SECTION 503—FIRE APPARATUS ACCESS ROADS

503.1 Where required. Fire apparatus access roads shall be provided and maintained in accordance with Sections 503.1.1 through 503.1.3.

C This section introduces the requirements for dedicated fire apparatus access roads serving new and relocated buildings in the jurisdiction. The requirements are to be established in coordination with the local fire service to accommodate the jurisdiction’s fire apparatus and equipment. The requirements are intended to provide the fire department with sufficient access to buildings to enable efficient fire suppression and rescue operations.

503.1.1 Buildings and facilities. *Approved* fire apparatus access roads shall be provided for every facility, building or portion of a building hereafter constructed or moved into or within the jurisdiction. The fire apparatus access road shall comply with the requirements of this section and shall extend to within 150 feet (45 720 mm) of all portions of the facility and all portions of the *exterior walls* of the first story of the building as measured by an *approved* route around the exterior of the building or facility.

Exceptions:

1. The *fire code official* is authorized to increase the dimension of 150 feet (45 720 mm) where any of the following conditions occur:
   - 1.1. The building is equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3.
   - 1.2. Fire apparatus access roads cannot be installed because of location on property, topography, waterways, nonnegotiable grades or other similar conditions, and an *approved* alternative means of fire protection is provided.
   - 1.3. There are not more than two Group R-3 or Group U occupancies.
2. Where *approved* by the *fire code official*, fire apparatus access roads shall be permitted to be exempted or modified for solar photovoltaic power generation facilities.

C This section establishes a requirement for a fire apparatus access road and the maximum distance from buildings or facilities to fire apparatus access roads. The provisions are intended to limit the maximum length of hose needed to reach any point along the exterior of a building or facility from a fire department vehicle. Large-area buildings may require a fire apparatus access road on all four sides. An access road is required to extend to within 150 feet (45 720 mm) of all portions along the exterior wall of the grade level story of each new or relocated building [see Commentary Figure 503.1.1(1)]. The 150-foot (45 720 mm) distance is based on the standard length of preconnected hoses carried on fire apparatus and is not intended to be measured to any point within the building.

A long, narrow building may require fire department access roads on two sides only, if all portions of the exterior of the grade level story are within 150 feet (45 720 mm) of the access road [see Commentary Figure 503.1.1(2)].

Small buildings may require an access road on one side only, if the access road is within 150 feet (45 720 mm) of all portions of the grade level floor [see Commentary Figure 503.1.1(3)].

In a case where new construction creates an addition ("... portion of a building hereafter constructed ...") to an existing building, the new building as completed should be evaluated as it relates to compliance with this section because the new addition increases the fuel load and potential firefighting complications. An addition to a building may require the construction of a new fire apparatus access road or the reconfiguration of an existing fire apparatus access road. This is consistent with the prescriptive compliance method provisions of Section 502.1 of the *International Existing Building Code®* (IEBC®), which require that the "... existing building or structure together with the addition are not less complying ... than the existing building or structure ... was prior to the addition." This is also consistent with IEBC Section 1101.2, which states, "An addition shall not create or extend any nonconformity in the existing building to which the addition is being made with regard to accessibility, structural strength, fire safety, means of egress, or the capacity of mechanical, plumbing, or electrical systems." In other words, an existing, nonconforming condition should not be made more nonconforming by construction of the addition. Since an addition is an increase in the area or height of an existing building, where a new building is erected immediately adjacent to an existing building, and they are separated by a fire wall constructed in accordance with Section 706 of the *International Building Code®* (IBC®), they are considered a separate building, not an addition to the existing structure. In that scenario, only the new building would be subject to this section. In any event, each case must be evaluated independently, especially if site topography or other conditions can slow or limit the response.

Condition 1.1 of Exception 1 states that the 150-foot (45 720 mm) distance may be increased, with the approval of the fire code official, where the building is equipped throughout with an NFPA 13, 13R or 13D automatic sprinkler system, as applicable. The code does not give the fire code official guidance on how much of an increase over 150 feet (45 720 mm) is reasonable. The fire code official must make the determination based on the response capabilities of his or her emergency response units and the anticipated magnitude of the incident.

The "alternative means" in Condition 1.2 of Exception 1 may include standpipes, automatic sprinklers, remote fire department connections or additional fire hydrants.

The Group R-3 facilities noted in Condition 1.3 of Exception 1 include one- and two-family dwellings and townhouses not falling within the scope of the *International Residential Code®* (IRC®); care facilities that accommodate five or fewer people; and congregate living facilities (nontransient) with 16 or fewer persons or congregate living facilities (transient) with 10 or fewer persons, among others. Group U occupancies are utility and miscellaneous accessory buildings or structures. Note that, since Section 903.2.8.1 requires that Group R-3 buildings be equipped throughout with automatic sprinkler systems, this condition is redundant with Condition 1.1. Note also that there is no exception for nonsprinklered buildings built under the IRC. See the commentary to IRC Section R101.2 and the commentary to IBC Section 310.1.

Exception 2 addresses photovoltaic panel system/array power generation facilities and provides guidance to jurisdictions in determining if a fire apparatus road is needed for hazard mitigation or if it can be exempted. Consideration must be given to the purpose of fire apparatus access roads within these facilities and how the provisions would be applied. Several issues arise when applying Section 503 of the code to ground-mounted photovoltaic systems/arrays. When considering the issues listed below, the fire code official should also consider other available code requirements that provide for appropriate hazard mitigation and risk reduction. Issues for consideration include:

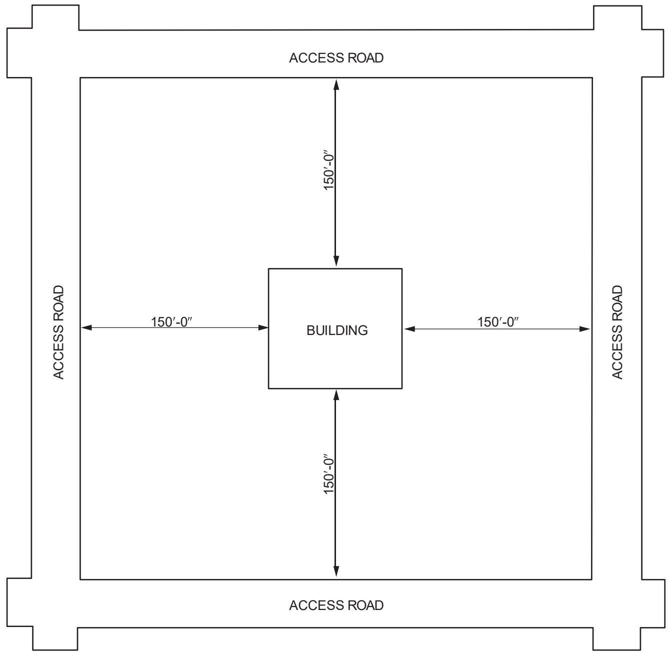
1. Risk/hazard to be mitigated.
2. Risk/hazard to firefighters or other emergency responders.
3. Interest of public safety and welfare.
4. Economics.
5. Intended access use.
6. Fuel load of the facility and adjacent areas that impact the facility.
7. Array configuration (tightly spaced, access aisles, height).
8. Actual hazard to public safety and welfare.

A question that often arises is whether code requirements pertaining to fire apparatus access roads are intended to be applicable to residential development sites upon which buildings are constructed under the provisions of the IRC. For information on this topic, see the commentary to Section 102.5.

Another question that arises is whether this section is intended to preclude locating a new building directly on a lot line, often referred to as a "zero lot line building." While it is true that some very large area buildings may require a fire apparatus access road on all four sides, this section does not specifically deal with exterior walls that may be located on the lot line in such buildings where the distance to a portion of that wall exceeds 150 feet (45 720 mm) from a fire apparatus access road. As such, the fire code official must determine the code's application in accordance with Section 102.9 and in consideration of the exception to this section.

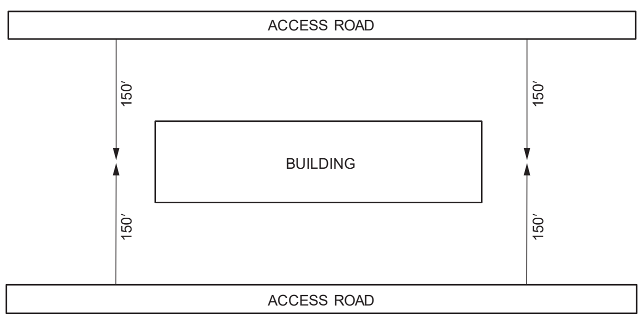
In determining the application, however, it should be considered that, in order for the fire department access contemplated by this section to be effective, an exterior wall would need to have openings in it through which access to the interior of the building could be achieved by hose streams or personnel. In the case of an exterior wall constructed on a property line with a zero-foot fire separation distance, IBC Table 705.5 requires that such walls have a fire-resistance rating of between 1 and 3 hours (depending on the occupancy group assigned to the building), and IBC Section and Table 705.9 require that such walls be without any openings. As such, access to the first (or any) floor level of that exterior wall would appear to provide little or no tactical usefulness to the fire department, especially if code-complying access is provided to other sides of the building.

Commentary Figure 503.1.1(1)—Fire Department Access—Large Building



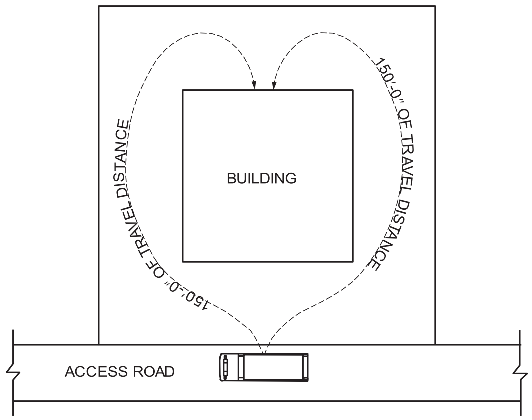
For SI: 1 foot = 304.8 mm.

Commentary Figure 503.1.1(2)—Fire Department Access on Two Sides



For SI: 1 foot = 304.8 mm.

Commentary Figure 503.1.1(3)—Small Building Access



For SI: 1 foot = 304.8 mm.

503.1.2 Additional access. The *fire code official* is authorized to require more than one fire apparatus access road based on the potential for impairment of a single road by vehicle congestion, condition of terrain, climatic conditions or other factors that could limit access.

C Additional access roads may be required by the fire code official based on his or her knowledge of traffic patterns, local weather conditions, terrain or the anticipated magnitude of a potential incident.

503.1.3 High-piled storage. Fire department vehicle access to buildings used for *high-piled combustible storage* shall comply with the applicable provisions of Chapter 32.

C Chapter 32 has special requirements for building access in occupancies with high-piled storage, but the requirements for fire apparatus access roads are the same as those required in this chapter.

503.2 Specifications. Fire apparatus access roads shall be installed and arranged in accordance with Sections 503.2.1 through 503.2.8.

C The dimensions of fire department access roads are based on the size, height and turning radius of emergency vehicles and the fact that emergency vehicles may be required to pass each other on the access road.

503.2.1 Dimensions. Fire apparatus access roads shall have an unobstructed width of not less than 20 feet (6096 mm), exclusive of shoulders, except for *approved* security gates in accordance with Section 503.6, and an unobstructed vertical clearance of not less than 13 feet 6 inches (4115 mm).

C The dimensions in this section are established to give fire apparatus continuous and unobstructed access to buildings and facilities.

This section requires that the unobstructed width of a fire apparatus access road must not be less than 20 feet (6096 mm). The intent of the minimum 20-foot (6096 mm) width is to provide space for fire apparatus to pass one another during fire-ground operations. The need to pass may occur when engines are parked for hydrant hook-up, laying hose or when trucks are performing aerial ladder operations. When an engine company is connected to a fire hydrant parallel to the curb using a front suction connection and using a side-discharge port on the pump, the horizontal distance that is needed to make a no-kink bend in the discharge fire hose can be considerable, especially when a large-diameter hose (LDH) is being used. The roadway width needed to accommodate such a common operational scenario would be the width of the apparatus plus the no-kink bending radius of the discharge hose, leaving minimal roadway width for other apparatus to squeeze by, if needed. Including adjacent road shoulders in the 20-foot (6096 mm) width measurement could yield substandard and inadequate driving surfaces for apparatus and, as such, shoulders are not to be included in the minimum width.

The minimum vertical clearance of 13 feet 6 inches (4115 mm) is the standard clearance used for highway bridges and underpasses. The vertical clearance requirement would apply in cases where a building or portion of a building, such as a canopy or porte-cochere, projects over all or a portion of the required width of the fire apparatus access road. Conversely, if the full required width of the fire apparatus access road is provided outside of the footprint of the projecting building element, the vertical clearance requirement would not apply. It is not intended that all projecting elements be constructed with a 13-foot 6-inch (4115 mm) vertical clearance, regardless of whether they encroach on the required width of a fire apparatus access road. Appendix D contains additional guidance on fire apparatus access road dimensions. It is important to note that the appendices are not considered part of the code unless specifically adopted.

503.2.2 Authority. The *fire code official* shall have the authority to require or permit modifications to the required access widths where they are inadequate for fire or rescue operations or where necessary to meet the public safety objectives of the jurisdiction.

C Fire departments respond to many types of emergency situations and the jurisdictions they serve may have traffic safety criteria that impact the design of access roadways used by emergency response vehicles. This section authorizes the fire code official to require greater, or to allow lesser, access-width dimensions based on the size and maneuverability of the anticipated emergency response apparatus, including mutual-aid apparatus from neighboring communities or agencies, among other considerations. This section also provides the flexibility needed for communities to address local issues or concerns. Types of communities include rural, suburban and urban, each presenting different issues and concerns, such as traffic safety, that may need to be addressed at a local level. It is difficult to anticipate every unique situation that may present itself and this section provides this flexibility.

503.2.3 Surface. Fire apparatus access roads shall be designed and maintained to support the imposed loads of fire apparatus and shall be surfaced so as to provide all-weather driving capabilities.

C This provision does not specify a particular type of surface. It is written in performance language; therefore, the surface must carry the load of the anticipated emergency response vehicles and be drivable in all kinds of weather.

The term “all-weather driving capabilities” would typically require some type of paved or hard surface. Gravel would be prone to problems in areas subject to heavy rain or in snowy climates where plowing could reduce the gravel roadbed to mud very quickly. Alternatives to concrete or asphalt, such as interlocking pavers, may be used where approved by the fire code official. Jurisdictions may benefit from developing a local policy outlining specific design requirements for fire apparatus access roads to clarify local interpretations of the section. The policy should include local requirements for surfacing and include acceptable surfacing materials.

503.2.4 Turning radius. The required turning radius of a fire apparatus access road shall be determined by the *fire code official*.

C The turning radius of an access road should be based on the turning radius of the anticipated responding emergency vehicles and must be approved by the fire code official.

503.2.5 Dead ends. Dead-end fire apparatus access roads in excess of 150 feet (45 720 mm) in length shall be provided with an *approved* area for turning around fire apparatus.

C In consideration of the hazards inherent in attempting to back emergency vehicles, especially larger ones such as tower ladders, out of a long dead-end roadway, this section is intended to create a safer situation by requiring that dead-end access roads over 150 feet long (45 720 mm) be equipped with an approved turnaround designed for the largest anticipated emergency-response vehicles. Appendix D contains examples of dead-end turnaround configurations. It is important to note that the appendices are not considered part of the code unless specifically adopted.

503.2.6 Bridges and elevated surfaces. Where a bridge or an elevated surface is part of a fire apparatus access road, the bridge shall be constructed and maintained in accordance with AASHTO HB-17. Bridges and elevated surfaces shall be designed for a live load sufficient to carry the imposed loads of fire apparatus. Vehicle load limits shall be posted at both entrances to bridges where required by the *fire code official*. Where elevated surfaces designed for emergency vehicle use are adjacent to surfaces that are not designed for such use, *approved* barriers, *approved* signs or both shall be installed and maintained where required by the *fire code official*.

C Bridges and elevated surfaces must be capable of carrying the weight of emergency response apparatus and must be marked with signage posting the weight limit of the bridge or elevated surface. Evaluation of bridges should be done in cooperation with the appropriate local or state agency having jurisdiction over private or public roadway bridges.

503.2.7 Grade. The grade of the fire apparatus access road shall be within the limits established by the *fire code official* based on the fire department’s apparatus.

C Generally, any grade exceeding 10 percent [e.g., greater than a 10-foot (3048 mm) rise in a 100-foot (30 480 mm) length] is required to have the approval of the fire code official. See Appendix D for additional guidance on fire apparatus access roads. Note that the appendices are not considered part of the code unless specifically adopted.

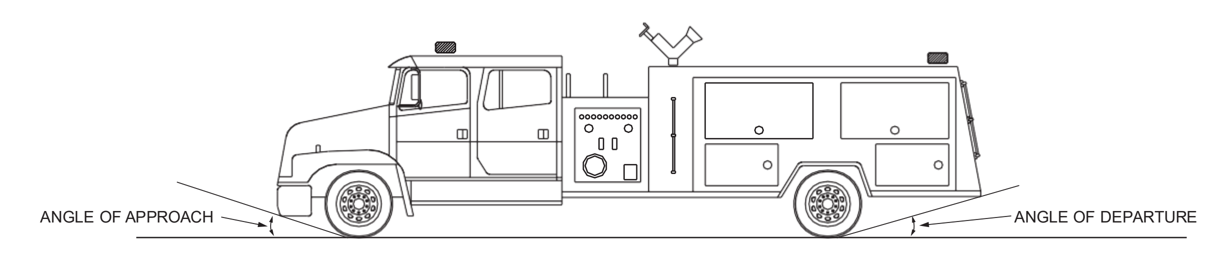
503.2.8 Angles of approach and departure. The angles of approach and departure for fire apparatus access roads shall be within the limits established by the *fire code official* based on the fire department’s apparatus.

C The angle of approach is the angle between the ground and a line running from the bottom of the front tire to the lowest-hanging point directly in front of it, which is usually the front bumper of the apparatus. This angle gives an indication of how steep an incline the vehicle can clear when approaching that angle.

The angle of departure is the angle between the ground and a line running from the bottom of the rear tire to the lowest-hanging point directly behind it, which is usually the rear step/tailboard of the apparatus. Similar to the approach angle, this angle indicates how steep an incline the vehicle can clear when departing from that angle.

These design aspects of a fire apparatus access road are crucial to successful navigation by apparatus and must be tailored to accommodate each piece of fire apparatus of the jurisdiction. See Commentary Figure 503.2.8.

Commentary Figure 503.2.8—Example of Angles of Approach and Departure



503.3 Marking. Where required by the *fire code official*, *approved* signs or other *approved* notices or markings that include the words "NO PARKING—FIRE LANE" shall be provided for fire apparatus access roads to identify such roads or prohibit the obstruction thereof. The means by which *fire lanes* are designated shall be maintained in a clean and legible condition at all times and be replaced or repaired when necessary to provide adequate visibility.

C Fire department access roads are normally designated on private property to provide fire service access; therefore, maintenance of the access roads, signage and any supplementary markings (pavement marking, curbs markings, etc.) are the responsibility of the owner of the property on which the fire apparatus road is located. Signage and supplemental markings should be in accordance with applicable local or state motor vehicle laws and should be enforced with the cooperation of the local police agency. Appendix D contains examples of signage. It is important to note that the appendices are not considered part of the code unless specifically adopted.

503.4 Obstruction of fire apparatus access roads. Fire apparatus access roads shall not be obstructed in any manner, including the parking of vehicles. The minimum widths and clearances established in Sections 503.2.1 and 503.2.2 shall be maintained at all times.

C To enforce "no parking" in fire apparatus access roads (fire lanes), the roads must be clearly marked. Some jurisdictions cite the building owner if the fire apparatus road is not properly marked and posted, and they cite the vehicle's owner for parking or blocking the access road if the access road is clearly marked. Other jurisdictions place the responsibility for marking the access roads, as well as the policing of "no parking" zones, on the building owner. In some states, motor vehicle laws may stipulate that fire apparatus access roads/fire lanes posted on private property may be enforced only by a traffic citation where an enforcement contract has been executed between the property owner and the local jurisdiction, that all markings be in accordance with the motor vehicle code and that the designated roadways be described in detail in the local "no parking" ordinances of the jurisdiction.

503.4.1 Traffic calming devices. Traffic calming devices shall be prohibited unless *approved* by the *fire code official*.

C This section prohibits the installation of traffic calming devices on fire apparatus access roads unless the devices are approved by the fire code official. What it doesn't do is detail how that approval is to be made within various jurisdictions. Each jurisdiction has its own traffic pattern emergency response challenges. The purpose of this requirement is to ensure that the fire department is part of this decision-making process. In most jurisdictions, the design and construction or review and approval of traffic calming devices is the responsibility of the municipal public works, transportation or engineering department. As a result, the fire code official and the appropriate engineering staff will need to work closely with one another to ensure that traffic calming devices, where approved, not only meet traffic engineering needs but also have the least impact on response time to emergencies. Traffic officials and fire code officials both have the responsibility to ensure that all public interests are properly considered in their decision-making process since both sets of officials have detailed regulations to provide for those interests. See the commentary to the definition of "Traffic calming device" in Section 202 for further discussion.

503.5 Required gates or barricades. The *fire code official* is authorized to require the installation and maintenance of gates or other *approved* barricades across fire apparatus access roads, trails or other accessways, not including public streets, alleys or highways. Electric gate operators, where provided, shall be *listed* in accordance with UL 325. Gates intended for automatic operation shall be designed, constructed and installed to comply with the requirements of ASTM F2200.

C The fire code official may require the installation and maintenance of gates or barricades across fire apparatus roads to prevent unauthorized vehicles from blocking or parking in the access road. The design and dimensions of the gates or barricades must be approved by the fire code official. Additionally, the gate or barricade must be operable or removable by the responding emergency units to provide them with quick access.

This section also addresses an important public safety issue regarding automatic operation of vehicular gates. Protection is needed from potential entrapment of individuals between an automatically moving gate and a stationary object or surface in close proximity to such gate. Gates intended for automation require specific design, construction and installation to accommodate entrapment protection to minimize or eliminate excessive gate gaps, openings and protrusions identified as contributing to the hazard of entrapments that have historically caused numerous serious injuries and deaths.

This section references two appropriate standards that deal with automatic gate safety: UL 325 and ASTM F2200. UL 325 is an ANSI-recognized safety standard containing provisions governing gate openers. Gate openers listed to the requirements of UL 325 provide an improved level of assurance that safety requirements have been met for such openers. ASTM F2200 is a consensus document containing provisions governing the construction of vehicular gates intended for automation and has been harmonized with the applicable provisions of UL 325.

503.5.1 Secured gates and barricades. Where required, gates and barricades shall be secured in an *approved* manner. Roads, trails and other accessways that have been closed and obstructed in the manner prescribed by Section 503.5 shall not be trespassed on or used unless authorized by the *owner* and the *fire code official*.

Exception: The restriction on use shall not apply to public officers acting within the scope of duty.

C The owner may secure the fire apparatus access road and restrict its use to emergency vehicles only, if warranted. If there is a need to secure gates or barricades, jurisdictions may require a padlock or key switch (on electrically operated gates) designed for the same key as key boxes provided in accordance with Section 506. Occasionally, electronically operated gates are required to be provided with a backup mechanism or breakaway feature in case the primary operating means fails.

The exception makes it clear that the securing of fire apparatus access roads is not intended to impede the legitimate and necessary use of the road by duly authorized public officers, such as police officers or other municipal personnel.

503.6 Security gates. The installation of security gates across a fire apparatus access road shall be *approved* by the *fire code official*. Where security gates are installed, they shall have an *approved* means of emergency operation. The security gates and the emergency operation shall be maintained operational at all times. Electric gate operators, where provided, shall be *listed* in accordance with UL 325. Gates intended for automatic operation shall be designed, constructed and installed to comply with the requirements of ASTM F2200.

C This section does not require that security gates be installed, but since they can affect fire department operations, their installation must be approved by the fire chief. Where installed, security gates must be operable in an emergency by the emergency response units and the means of operation must be acceptable to the fire chief. Electrically operated gates should also include a manual method of operation (also see commentary, Section 503.5.1).

This section requires ongoing maintenance of security gates so that ready access to the roadway may be accomplished. If gates are not maintained in a manner that prevents appreciable delay of emergency response, the fire code official has the authority to have gates removed because they would be considered an obstruction of the required roadway width as regulated in Section 503.4.

This section also addresses an important public safety issue regarding automatic operation of vehicular gates. Protection is needed from potential entrapment of individuals between an automatically moving gate and a stationary object or surface in close proximity to such gate. Gates intended for automation require specific design, construction and installation to accommodate entrapment protection to minimize or eliminate excessive gate gaps, openings and protrusions identified as contributing to the hazard of entrapments that have historically caused numerous serious injuries and deaths.

This section references two appropriate standards that deal with automatic gate safety: UL 325 and ASTM F2200. UL 325 is an ANSI-recognized safety standard containing provisions governing gate openers. Gate openers listed to the requirements of UL 325 provide an improved level of assurance that safety requirements have been met for such openers. ASTM F2200 is a consensus document containing provisions governing the construction of vehicular gates intended for automation and has been harmonized with the applicable provisions of UL 325.

SECTION 504—ACCESS TO BUILDING OPENINGS AND ROOFS

504.1 Required access. Exterior doors and openings required by this code or the *International Building Code* shall be maintained with ready access for emergency access by the fire department. An *approved* access walkway leading from fire apparatus access roads to exterior openings shall be provided where required by the *fire code official*.

C The exterior openings referred to in this section are typically exit discharge doors, since such openings provide fire department access directly into the building or to a fire-resistance-rated enclosure from which to operate in multiple-story buildings. This section also includes access openings for rack or high-piled storage buildings required by Section 3206.7. Under certain circumstances, in order for emergency response personnel to get equipment from the emergency apparatus to the building, the fire code official is authorized to require approved walkways from the apparatus access road to the building openings on grade level.

504.2 Maintenance of exterior doors and openings. Exterior doors and their function shall not be eliminated without prior approval. Exterior doors that have been rendered nonfunctional and that retain a functional door exterior appearance shall have a sign affixed to the exterior side of the door with the words “THIS DOOR BLOCKED.” The sign shall consist of letters having a principal stroke of not less than $^{3}/_{4}$ inch (19.1 mm) wide and not less than 6 inches (152 mm) high on a contrasting background. Required fire department access doors shall not be obstructed or eliminated. *Exit* and *exit access* doors shall comply with Chapter 10. Access doors for *high-piled combustible storage* shall comply with Section 3206.7.

C This section pertains to emergency access openings and to all exterior doors that have a functional appearance from the outside but are not operable. Doors that are part of the required means of egress or that are required by Section 3206.7 must not be rendered unusable or blocked. Only doors not required by the code or the IBC for means of egress may be blocked or made unusable. Even then, they must be marked from the outside so that emergency personnel will not attempt to use them.

504.3 Stairway access to roof. New buildings four or more stories above grade plane, except those with a roof slope greater than 4 units vertical in 12 units horizontal (33.3 percent slope), shall be provided with a *stairway* to the roof. *Stairway* access to the roof shall be in accordance with Section 1011.12. Such *stairway* shall be marked at street and floor levels with a sign indicating that the *stairway* continues to the roof. Where the roof is a *vegetative roof*, includes a *landscaped roof* area, or is used or for other purposes, stairways shall be provided as required for such occupancy classification.

C The stairway to the roof, required by this section, must comply with Sections 1011.12 and 1011.12.2. If the stairway to the roof serves as a means of egress from an occupiable roof, then the stairway must be equipped with all the components of an exit stairway, such as the required riser and tread dimensions, handrails, etc. If the stairway to the roof is not required for roof egress, Section 1011.14 allows the stairway segment from the top floor to the roof to be an alternating tread device. The access to the roof is required for firefighter use and not the general public; therefore, the door leading to the roof may be secured in a manner approved by the fire code official with due consideration given to whether the door is an egress element for the roof. These provisions apply only to new buildings four or more stories in height (see commentaries, Sections 1011.12 and 1023.9).

SECTION 505—PREMISES IDENTIFICATION

505.1 Address identification. New and existing buildings shall be provided with *approved* address identification. The address identification shall be legible and placed in a position that is visible from the street or road fronting the property. Address identification characters shall contrast with their background. Address numbers shall be Arabic numbers or alphabetical letters. Numbers shall not be spelled out. Each character shall be not less than 4 inches (102 mm) high with a minimum stroke width of $^{1}/_{2}$ inch (12.7 mm). Where required by the *fire code official*, address identification shall be provided in additional *approved* locations to facilitate emergency response. Where access is by means of a private road and the building cannot be viewed from the *public way*, a monument, pole or other sign or means shall be used to identify the structure. Address identification shall be maintained.

C Buildings must be provided and maintained with address identification that is easily identifiable by emergency responders from the emergency response vehicle. This should include the backs of buildings that face alleys or roads, since the emergency response unit may often be directed to the back entrance to a building, such as in a strip shopping center. The back door of each tenant space should have the numerical address and store name on or above the door.

This section also provides the fire code official with the authority to require additional address locations for facilities with unusual or problematic configurations, such as college or hospital campuses, strip malls, business parks, apartment complexes and other complex properties where identification of buildings is essential to emergency responders. The additional requirements proposed by the added language will assist various emergency responders in identifying specific addresses when an emergency response from locations other than the primary access point is required.

There are also situations where a building’s setback from the street is so large that the building itself may not even be visible to emergency responders. In such circumstances, this section requires that the road or driveway giving access to the building be marked to assist emergency responders in promptly identifying their access. It is important that such remote markings be reviewed and approved by the fire code official prior to installation.

505.2 Street or road signs. Streets and roads shall be identified with *approved* signs. Temporary signs shall be installed at each street intersection when construction of new roadways allows passage by vehicles. Signs shall be of an *approved* size, weather resistant and be maintained until replaced by permanent signs.

C The names of streets in new developments may not be on maps, making them hard for emergency responders to find. Temporary street signs must be installed before construction begins and replaced later with permanent signs.

SECTION 506—KEY BOXES

506.1 Where required. Where access to or within a structure or an area is restricted because of secured openings or where immediate access is necessary for life-saving or firefighting purposes, the *fire code official* is authorized to require a key box to be installed in an *approved* location. The key box shall be of an *approved* type *listed* in accordance with UL 1037, and shall contain keys to gain necessary access as required by the *fire code official*.

C The fire code official has the authority to require special key vaults where, in his or her opinion, the need for rapid entry into facilities warrants it. The key boxes or vaults are located on the exterior of the building for ready access and are openable with a special master key in the possession of the emergency responders. See Commentary Figures 506.1(1) and 506.1(2) and the commentaries to the definitions of "Key box" and "Fire department master key" in Section 202.

This section also specifies a level of security for key boxes. Where a rapid-entry key box is required, there is an obligation to make sure that the key box required or approved by the fire code official is secure to prevent the key box from becoming a security threat. Security is addressed by requiring an approved key box to be listed in accordance with UL 1037. The major key box manufacturers have their rapid entry devices listed under this standard.

This section is a discretionary provision based on the jurisdiction's determination that the use of a key box is safer for firefighters than performing manual forcible entry into a structure. If properly maintained in accordance with Section 506.2, a key box can also lower property losses because it expedites the entry of firefighters into a structure without necessitating the damage associated with forcing entry. The decision for the use of key boxes rests solely with the fire code official.

Commentary Figure 506.1(1)—Surface-Mounted Key Box

Photo courtesy of Knox Company

Commentary Figure 506.1(2)—Recessed-Mounted Key Box

Photo courtesy of Knox Company

506.1.1 Locks. An *approved* lock shall be installed on gates or similar barriers where required by the *fire code official*.

C The key box suppliers also have special padlocks and electronic key-operated switches that are controlled by the same fire department master key that opens the key vaults. These padlocks can be required by the fire code official for security gates (also see commentaries, Sections 503.5, 503.5.1 and 503.6). The key-activated electronic switches may also be required for the control of certain equipment in the building, such as smoke control equipment, or to shut down a dangerous process.

506.1.2 Key boxes for nonstandardized fire service elevator keys. Key boxes provided for nonstandardized fire service elevator keys shall comply with Section 506.1 and all of the following:

1. The key box shall be compatible with an existing rapid entry key box system in use in the jurisdiction and *approved* by the *fire code official*.
2. The front cover shall be permanently labeled with the words "FIRE DEPARTMENT USE ONLY—ELEVATOR KEYS."
3. The key box shall be mounted at each elevator bank at the lobby nearest to the lowest level of fire department access.
4. The key box shall be mounted 5 feet 6 inches (1676 mm) above the finished floor to the right side of the elevator bank.
5. Contents of the key box are limited to fire service elevator keys. Additional elevator access tools, keys and information pertinent to emergency planning or elevator access shall be permitted where authorized by the *fire code official*.
6. In buildings with two or more elevator banks, a single key box shall be permitted to be used where such elevator banks are separated by not more than 30 feet (9144 mm). Additional key boxes shall be provided for each individual elevator or elevator bank separated by more than 30 feet (9144 mm).

Exception: A single key box shall be permitted to be located adjacent to a *fire command center* or the nonstandard fire service elevator key shall be permitted to be secured in a key box used for other purposes and located in accordance with Section 506.1.

C Many jurisdictions have elevators built prior to the introduction of a standardized elevator key and, as a result, each building with an elevator requires its own elevator key. The key is most likely to be based on the model of the elevator and the year it was manufactured. This section sets forth requirements to assist jurisdictions in managing the issue of having different elevator keys in different buildings. These provisions specify where the key box is to be located inside the building and require it to be compatible with the rapid-entry existing key boxes that may already be in use in the jurisdiction. The exception allows key box installation adjacent to a fire command center or in other locations where approved by the fire code official. See the commentaries to Sections 604.6 through 604.6.2.4 for a discussion of standardized fire service elevator keys.

506.2 Key box maintenance. The operator of the building shall immediately notify the *fire code official* and provide the new key where a lock is changed or rekeyed. The key to such lock shall be secured in the key box.

C In most cases, the owner of a building cannot open the key vault or box and must call the fire code official to have someone open it to replace keys that have been changed. The building owner is responsible for maintaining the key box, as well as keeping the keys inside the box current.

SECTION 507—FIRE PROTECTION WATER SUPPLIES

507.1 Required water supply. An *approved* water supply capable of supplying the required fire flow for fire protection shall be provided to premises on which facilities, buildings or portions of buildings are hereafter constructed or moved into or within the jurisdiction.

C This section requires that adequate fire protection water be provided to premises on which new buildings are constructed or onto which a building is moved, from either outside of the jurisdiction or another location within the jurisdiction. Note that this section states that the water supply must be capable of supplying the required fire flow to the premises; however, the means by which the fire flow is supplied is determined by the policies of the jurisdiction, such as a pumper taking suction from a hydrant, tanker or lake (also see Appendices B and C for further information on fire flows and fire hydrants). It is important to note that the appendices are not considered part of the code unless specifically adopted. The phrase "... hereafter constructed or moved ..." used in this section (and in Appendices B and C, if duly adopted) limits the application of water supply provisions to only newly erected or relocated buildings, as opposed to existing buildings or existing remodeled buildings.

A question that often arises is whether the code's regulations pertaining to fire protection water supply are intended to be applicable to one- and two-family residential development sites on which buildings are constructed under the provisions of the IRC. The IRC is intended to be a stand-alone code for the construction of detached one- and two-family dwellings and townhouses not more than three stories in height. As such, all of the provisions for the construction of buildings of those descriptions are to be regulated exclusively by the IRC and not by another I-Code. However, the IRC applies only to the construction of the structures of those buildings and not to the development of the site on which such structures are built. Accordingly, where this code is adopted, its fire protection water supply provisions (including specifically adopted related appendices) apply. This code's requirements address only land development requirements for providing fire protection water supply to residential sites on the same basis as to the rest of the community (also see commentaries, Sections 102.5 and 503.1.1).

507.2 Type of water supply. A water supply shall consist of reservoirs, pressure tanks, elevated tanks, water mains or other fixed systems capable of providing the required fire flow.

C A good water supply consists of an adequate source of water, distribution system and proper pressure for delivery. If the water source is not reliable, it should not be considered as an acceptable water supply.

507.2.1 Private fire service mains. Private fire service mains and appurtenances shall be installed in accordance with NFPA 24.

C Private fire service mains are often installed on private property where facilities are located well away from municipal water distribution systems. Private hydrants likely cannot be installed on mains less than 6 inches (152 mm) in diameter (see Section 5.2.1.2 of NFPA 24). Essentially, the hydraulic calculations are necessary to show the main is able to supply the total demand at the appropriate pressure. This is not likely with pipe sizes less than 6 inches (152 mm). However, such mains have other possible applications such as for automatic sprinkler systems. Where installed, private fire service mains, private hydrants, control valves, hose houses and related equipment must be installed and maintained in accordance with NFPA 24. Private (yard) hydrants may be tested, painted and marked in accordance with NFPA 291 where approved by the fire code official and the fire department.

507.2.2 Water tanks. Water tanks for private fire protection shall be installed in accordance with NFPA 22.

C Water tanks for private fire protection may be required where municipal water systems do not exist or are incapable of supplying sprinkler or standpipe demand, or where IBC Section 403.3.3 requires a secondary water supply for high-rise buildings in Seismic Design Category C, D, E or F. NFPA 22 and IBC Section 1511.3 govern the installation of water tanks on buildings. Pressure tanks must bear the label of an approved agency and be installed in accordance with the manufacturer's instructions.

507.3 Fire flow. Fire-flow requirements for buildings or portions of buildings and facilities shall be determined by an *approved* method.

C Appendix B of the code, which sets forth minimum fire flow requirements for one- and two-family dwellings and commercial buildings, offers one method for determining fire flow and its duration that could be approved by the fire code official. In areas that do not have a water supply, such as rural areas with no conventional water storage and distribution system, the jurisdiction may choose to utilize the methods contained in NFPA 1142, *Standard on Water Supplies for Suburban and*

Rural Firefighting; the *International Wildland-Urban Interface Code*® (IWUIC®); or the ISO *Guide for Determination of Needed Fire Flow*. Appendix Table B105.1(2) bases fire flow on the type of construction and the square footage of the fire flow calculation area. All calculations in the table are based on a 20-pounds-per-square-inch (psi) (138 kPa) residual pressure. Note that the provisions of Section B103 provide for increases, reductions and specific alternative methods for determining flows. In addition to Section B103, Sections 104.2.3 and 104.2.4 also provide the fire code official with authority concerning modifications and alternative methods. See the commentary to Appendix B for discussion of the fire flow requirements. It is important to note that the appendices are not considered part of the code unless specifically adopted.

507.4 Water supply test. The *fire code official* shall be notified prior to the water supply test. Water supply tests shall be witnessed by the *fire code official* or *approved* documentation of the test shall be provided to the *fire code official* prior to final approval of the water supply system.

C Whatever type of water supply is proposed in order to comply with Section 507.1, it must be tested in a manner that will verify it is capable of providing the required fire flow. The water supply system must be tested, and the contractor is required to notify the fire code official prior to performing the test on the system. The fire code official will make the final approval by either witnessing the test or accepting the certification documentation. NFPA 291 contains recommended test methodology.

If the water supply system includes private fire service mains, NFPA 24, referenced in Section 507.2.1, contains the testing requirements for private fire service mains, as well as a test certificate form. It should be noted that the test certificate form has signature blocks only for the building owner's representative and the installing contractor's representative. There is no place on the form for the fire code official's signature, nor should he or she expose him or herself to liability of any kind for the installation by signing the form.

507.5 Fire hydrant systems. Fire hydrant systems shall comply with Sections 507.5.1 through 507.5.6.

C Where fire hydrant systems are part of the approved water supply, the system must comply with this section.

507.5.1 Where required. Where a portion of the facility or building hereafter constructed or moved into or within the jurisdiction is more than 400 feet (122 m) from a hydrant on a fire apparatus access road, as measured by an *approved* route around the exterior of the facility or building, on-site fire hydrants and mains shall be provided where required by the *fire code official*.

Exceptions:

1. For Group R-3 and Group U occupancies, the distance requirement shall be 600 feet (183 m).
2. For buildings equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2, the distance requirement shall be 600 feet (183 m).

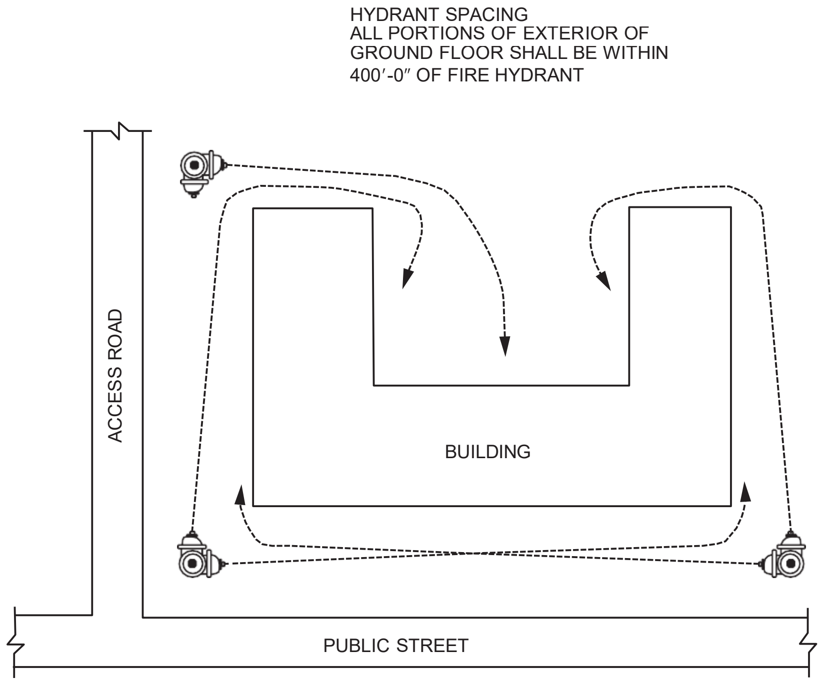
C The intent is that not more than 400 feet (122 m) of hose will have to be laid out to reach all portions of the exterior grade level of the building. Each hydrant must be accessible to fire apparatus and the 400-foot (122 m) distance should be measured from the hydrant(s) to all portions of the exterior at ground level [see Commentary Figure 507.5.1(1)]. When on-site hydrants and mains are required to achieve compliance with the distance requirement, this section gives the fire code official the authority to determine and approve hydrant locations on the site.

This paragraph is not intended to prevent development in rural areas where fire hydrants are not available, as long as the fire code official has approved an alternative water supply. The alternative water supply could be a fire department water tanker or a static, all-weather water supply that is approved by the fire code official.

In recognition of the smaller relative size and fire hazard characteristics of one- and two-family dwellings and utility buildings, Exception 1 increases the 400-foot (122 m) distance to 600 feet (183 m) [see Commentary Figure 507.5.1(2)]. Note that the one- and two-family dwellings classified in Group R-3 are those that are not within the scope of the IRC. See Section 102.5.

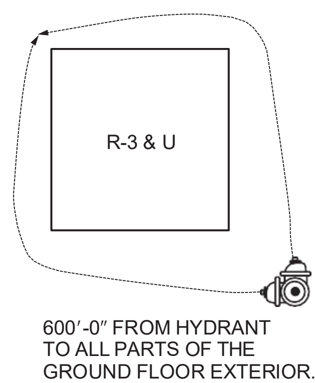
In recognition of the proven efficiency of sprinklers in applying water directly on the seat of the fire for buildings equipped throughout with automatic sprinklers in accordance with NFPA 13 or NFPA 13R, as applicable, Exception 2 increases the 400-foot (122 m) distance to 600 feet (183 m) [see Commentary Figure 507.5.1(3)].

Commentary Figure 507.5.1(1)—Hydrant Layout to Minimize Hose Run



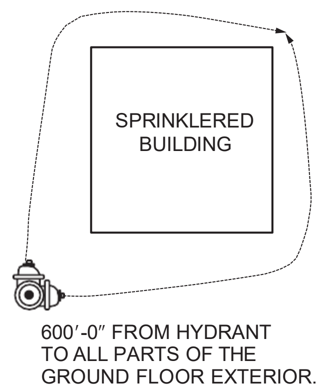
For SI: 1 foot = 304.8.

Commentary Figure 507.5.1(2)—Hydrant Location—Groups R-3 and U Occupancies



For SI: 1 foot = 304.8.

Commentary Figure 507.5.1(3)—Acceptable Hydrant Location with Sprinklers



For SI: 1 foot = 304.8.

507.5.1.1 Hydrant for standpipe systems. Buildings equipped with a standpipe system installed in accordance with Section 905 shall have a fire hydrant within 100 feet (30 480 mm) of the fire department connections.

Exception: The distance shall be permitted to exceed 100 feet (30 480 mm) where *approved* by the *fire code official*.

C This section provides correlation with NFPA 14, Section 9.9.5.4 (2024 edition), which requires that standpipe fire department connections be placed within 100 feet (30 m) of a fire hydrant, unless otherwise approved by the authority having jurisdiction. However, that requirement is frequently missed where site work and approval has been based only on the fire apparatus access road and fire protection water supply requirements in this chapter. With the requirement now included in this section, the code user is directed to this requirement during the site design review stage and not as an afterthought as has often happened during the building permit site plan review. The exception provides design flexibility but protects operational needs of the fire department by requiring the fire code official to approve any increase in distance between the fire department connection (FDC) and the hydrant. Note that this requirement is not applicable to sprinkler connections since NFPA 13 does not have a distance requirement to a fire hydrant for connections serving only sprinkler systems.

507.5.2 Inspection, testing and maintenance. Fire hydrant systems shall be subject to periodic tests as required by the *fire code official*. Fire hydrant systems shall be maintained in an operative condition at all times and shall be repaired where defective. Additions, repairs, *alterations* and servicing shall comply with *approved* standards. Records of tests and required maintenance shall be maintained.

C The fire code official has the authority to require periodic tests and to specify the frequency of such tests. The generally accepted procedure is to inspect hydrants annually for proper operation and drainage by opening and closing the hydrants and lubricating all threads. This section also requires that written inspection and maintenance records be kept. Such records should indicate the date and time and the name of the person conducting the inspection or maintenance. These records must be maintained by the owner and should be made available to the fire code official for review when requested. This requirement relieves the fire code official of the administrative burden of maintaining test records.

507.5.3 Private fire service mains and water tanks. Private fire service mains and water tanks shall be periodically inspected, tested and maintained in accordance with NFPA 25 at the following intervals:

1. Private fire hydrants of all types: Inspection annually and after each operation; flow test and maintenance annually.
2. Fire service main piping: Inspection of exposed, annually; flow test every 5 years.
3. Fire service main piping strainers: Inspection and maintenance after each use.

Records of inspections, testing and maintenance shall be maintained.

C NFPA 25 is the *Standard for the Inspection, Testing and Maintenance of Water-Based Fire Protection Systems*. Chapter 7 of that standard covers private fire service mains and Chapter 9 covers water storage tanks. This section also requires that written inspection and maintenance records be kept. Such records should indicate the date and time and the name of the person conducting the inspection or maintenance. These records must be maintained by the owner and should be made available to the fire code official for review when requested. This requirement relieves the fire code official of the administrative burden of maintaining test records.

507.5.4 Obstruction. Unobstructed access to fire hydrants shall be maintained at all times. The fire department shall not be deterred or hindered from gaining immediate access to fire protection equipment or fire hydrants.

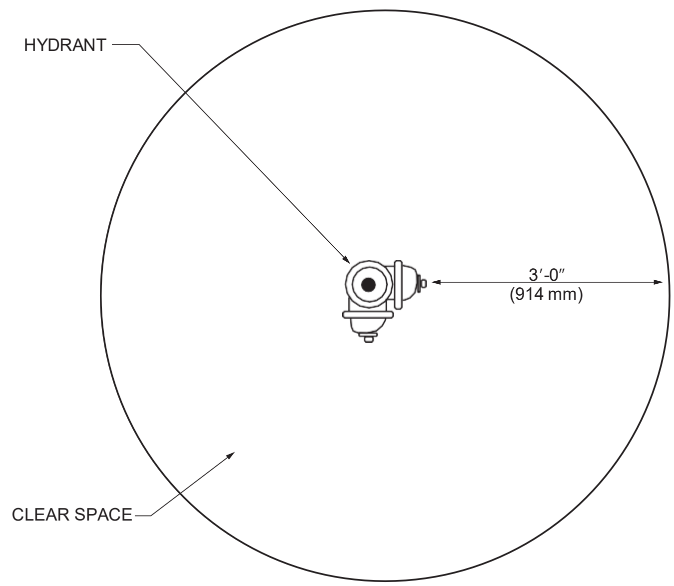
C Nothing can be allowed to be placed near a fire hydrant, FDC or control valve that would prevent responding firefighters from immediately recognizing the device and gaining access to it. Plants and shrubs on public or private property are probably the most common object that can make fire hydrants, FDCs or fire protection system valves virtually invisible to responding fire apparatus engineers. In residential areas especially, some homeowners do not like “that ugly piece of iron” (i.e., a fire hydrant) in their yard, so they plant all manner of vegetation around it in an effort to hide it—a clear violation of this section. On construction sites, fire hydrants or FDCs are often hidden from view and access by deliveries of construction materials randomly dumped at the most convenient spot on the site without regard for the need of the fire department to gain immediate access to hydrants or FDCs.

507.5.5 Clear space around hydrants. A 3-foot (914 mm) clear space shall be maintained around the circumference of fire hydrants, except as otherwise required or *approved*.

C Care must be taken so that fences, utility poles, barricades and other obstructions do not prevent access to and operation of fire hydrants. A clear space of 3 feet (914 mm) must be maintained around hydrants (see Commentary Figure 507.5.5) to allow easy hose connections to the hydrant and the efficient use of hydrant wrenches, spanner wrenches and other tools needed by the apparatus engineer. This section is not intended to allow any of the obstructions described in Section 507.5.4 to exist as long as they are kept 3 feet (914 mm) from the hydrant, FDC or valve.

Though not specifically mentioned in this section, it is also important that hydrants be installed with the center of the outlet cap nuts at least 18 inches (457 mm) above adjoining grade to accommodate the free turning of a hydrant wrench when removing the caps (see NFPA 24, Chapter 7, for further information).

Commentary Figure 507.5.5—Hydrant Clearance

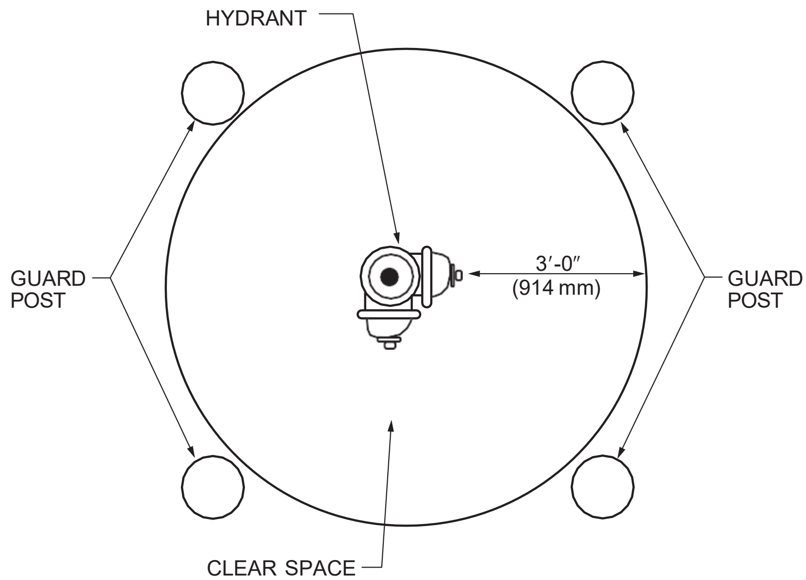


For SI: 1 foot = 304.8 mm.

507.5.6 Physical protection. Where fire hydrants are subject to impact by a motor vehicle, guard posts or other *approved* means shall comply with Section 312.

C Section 312 requires vehicle impact protection by placing steel posts filled with concrete around the hydrant (see Commentary Figure 507.5.6). Section 312 gives the specifications for the posts. Note that the provisions of Section 507.5.5 apply to the installation of posts or other protective features.

Commentary Figure 507.5.6—Hydrant Impact Protection



For SI: 1 foot = 304.8 mm.

SECTION 508—FIRE COMMAND CENTER

508.1 General. Where required by other sections of this code and in all buildings classified as high-rise buildings by the *International Building Code* and in all F-1 and S-1 occupancies with a building footprint greater than 500,000 square feet (46 452 m^2), a *fire command center* for fire department operations shall be provided and shall comply with Sections 508.1.1 through 508.1.7.

C Fire ground operations usually involve establishing an incident command post where the incident command officer can observe what is happening, control arriving personnel and equipment, and direct the resources and firefighting operations effectively. A fire command center provides a centralized location for the incident commander to review details about the building and the incident and to effectively coordinate emergency responders and suppression activities with increased efficiency and speed.

There are three overall situations where fire command centers are required:

High-rise buildings. Because of the difficulties in controlling a fire in a high-rise building, an adequately sized, protected and easy-to-access separate room for this purpose within the building must be established to assist the incident command officer (see also Section 202 commentary to the definition of “Fire command center”).

Smoke-protected seating. A fire command center is required by Section 909.16 in buildings containing smoke-protected assembly seating to house the firefighter’s smoke control panel. Facilities with smoke-protected seating tend to be larger facilities that, at the very least, would already have a central security center that could also function as a fire command center where approved by the jurisdiction (see commentary, Section 909.16). Note that the smoke-protected seating requirements are found in Section 1030.6.

Group F-1 and S-1 occupancies. Group F-1 and S-1 buildings in excess of 500,000 square feet (46 452 m^2) can be just as challenging for firefighters as high-rise buildings. A fire command center in these types of buildings will allow the incident commander to see at a glance, in one protected location, where the fire is, the building layout and any active fire protection to provide the best strategy possible to mitigate the problem and protect the lives of firefighters. To put into context, a fire command center would be triggered where the buildings are approximately five times the typical warehouse store and approximately three times the size of a typical retail supercenter. Large structures of this size pose numerous challenges to emergency responders due to the large amounts of fuel loads from the storage, manufacturing and/or processing of flammable/combustible commodities and other processes within the building. Challenges include wide distribution of smoke throughout the structure; difficulty for firefighters to locate and reach the fire; and difficulty in search and evacuation of the public, employees and firefighters. These structures typically require numerous fire protection measures such as early suppression and detection systems that may include, but are not limited to, fire pumps, multiple fire sprinkler systems, advanced fire alarm systems, smoke control systems and refrigeration gas detection system(s). During a fire, the incident commander must have the ability to readily identify the status of the various suppression and detection systems. In addition, the incident commander must have access to other building information details that may include, but are not limited to, building floor plans, high-pile/rack storage details, smoke control/ventilation systems, fire sprinkler zoning details, mechanical refrigeration equipment and piping details, and hazardous materials data sheets along with quantities and storage/use locations.

508.1.1 Location and access. The location and access to the *fire command center* shall be *approved* by the *fire code official*.

C Because of its importance to fire suppression and rescue operations, the fire command center must be provided at a location that is acceptable to the fire department, usually near the front of the building near the main entrance, so that the first arriving command officer can access it quickly and undertake operations. Though this section references the fire code official, in many cases the command officer and fire code official may be one and the same. In cases where they are not, it is expected that input from those who will fight fires and respond to emergencies will be accounted for in this decision.

508.1.2 Separation. The *fire command center* shall be separated from the remainder of the building by not less than a 1-hour *fire barrier* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assembly* constructed in accordance with Section 711 of the *International Building Code*, or both.

C Because of its importance to fire suppression and rescue operations, the fire command center must be separated from the remainder of the building by 1-hour fire barriers and horizontal assemblies, including opening protectives, to protect the room, its contents and occupants from an incident in adjacent areas of the building, and to limit noise and distractions during command operations within the room.

508.1.3 Size. The *fire command center* shall be not less than 0.015 percent of the total building area of the facility served or 200 square feet (19 m^2) in area, whichever is greater, with a minimum dimension of 0.7 times the square root of the room area or 10 feet (3048 mm), whichever is greater.

Where a *fire command center* is required for Group F-1 and S-1 occupancies with a building footprint greater than 500,000 square feet (46 452 m^2), the *fire command center* shall have a minimum size of 96 square feet (9 m^2) with a minimum dimension of 8 feet (2438 mm) where *approved* by the *fire code official.*

C This section is intended to provide a minimum size and configuration of the fire command center that allows sufficient space for the necessary command personnel to effectively perform required tasks associated with a fire command center without interfering with each other. Fire command centers must be designed to accommodate several emergency response commanders wearing full protective equipment, provide space to review building emergency plans during incidents, co-locate decision makers within the incident command system (ICS), and interpret fire protection system and building system information generated by the features required by Section 508.1.6.

The sizing requirements are addressed in two different ways. The first applies to high-rise buildings and assembly occupancies utilizing smoke-protected seating. The second method focuses on very large Group F-1 and S-1 occupancies.

For high-rise buildings and buildings applying the smoke-protected seating option using mechanical smoke control, this section recognizes that larger buildings may need a larger fire command center due to the increase in complexity of systems and associate space needs they may create. Therefore, criteria are provided to determine what size the fire command center must be. For these buildings, the fire command center must be a minimum of 200 square feet (19 m^2) or 0.015 percent of the total building area, whichever is greater. This percentage is fairly low and would not amount to an increase of over 200 square feet (19 m^2) until a building reaches 1.3 million square feet in building area. The following equation shows how the minimum building square footage is derived:

Total building area × 0.015/100 = 200 square feet minimum

Total building area = 200 square feet × 100/0.015

Total building area = 1,333,333 square feet (1.3 million square feet)

The criteria for a minimum width are also provided as 0.7 times the square root of the room area or 10 feet (3048 mm), whichever is greater. In most cases, the width would be 10 feet (3048 mm). The increase in width is necessary for rooms with the following area or greater. The following equation shows the maximum square footage where the width of the fire command center can remain at 10 feet (3048 mm). Anything greater than this size would require an increased width.

$$\sqrt{\text{Area of the fire command center}} \times 0.7 = 10 \text{ feet}$$

$$\sqrt{\text{Area of the fire command center}} = 10 \text{ feet}/0.7$$

$$\text{Area of the fire command center} = \left(\frac{10 \text{ feet}}{0.7}\right)^2$$

Area of the fire command center = 204 square feet

Given the multiple uses of a fire command center, any smaller room would compromise the effectiveness of incident management for high-rise buildings or assembly occupancies utilizing smoke-protected seating with very large building areas.

For very large Group F-1 and S-1 occupancies, the need for a fire command center is more critical than the size. The minimum size need only be 96 square feet (9 m^2) with a more limited width of 8 feet (2438 mm). No increases in size are required as the building gets larger.

508.1.4 Layout approval. A layout of the *fire command center* and all features required by this section to be contained therein shall be submitted for approval prior to installation.

C The flow of critical tactical information into, within and out of a fire command center is, by its very nature, both high in volume and intense in nature, and has a direct bearing on the safety of building occupants and emergency response personnel at work at an incident. For that reason, the layout and arrangement of the fire command center must comport with the operational procedures of the local fire department to optimize the receipt, processing and dissemination of operational information and orders. Accordingly, the fire code official must review and approve the arrangement of the fire command center prior to the installation of any of the controls and features required by Section 508.1.6. Given the operational importance of the fire command center, the fire code official should work closely with the jurisdiction's fire chief to make sure that all operational needs are identified and met during the design stages.

508.1.5 Storage. Storage unrelated to operation of the *fire command center* shall be prohibited.

C Fire command centers are unique rooms in unique buildings and are strictly reserved for emergency management operations. As such, they must be neat and orderly at all times so as not to obstruct or limit access to all of the system controls they contain. This section supports that need by prohibiting the storage within a fire command center of anything not directly related to the function of fire command.

508.1.6 Required features. The *fire command center* shall comply with NFPA 72 and shall contain the following features:

1. The emergency voice/alarm communications system control unit.
2. The fire department communications system.
3. Fire detection and alarm system annunciator.
4. Annunciator unit visually indicating the location of the elevators and whether they are operational.
5. Status indicators and controls for air distribution systems.
6. The firefighter's control panel required by Section 909.16 for smoke control systems installed in the building.
7. Controls for unlocking *interior exit stairway* doors simultaneously.
8. Sprinkler valve and water-flow detector display panels.
9. Emergency and standby power status indicators.
10. A telephone for fire department use with controlled access to the public telephone system.
11. Fire pump status indicators.
12. Schematic building plans indicating the typical floor plan and detailing the building core, *means of egress*, *fire protection systems*, firefighter air-replenishment systems, firefighting equipment and fire department access, and the location of *fire walls*, *fire barriers, fire partitions, smoke barriers* and smoke partitions.
13. An *approved* Building Information Card that includes, but is not limited to, all of the following information:
    - 13.1. General building information that includes: property name, address, the number of floors in the building above and below grade, use and occupancy classification (for mixed uses, identify the different types of occupancies on each floor) and the estimated building population during the day, night and weekend.
    - 13.2. Building emergency contact information that includes: a list of the building's emergency contacts including but not limited to building manager, building engineer and their respective work phone number, cell phone number and email address.
    - 13.3. Building construction information that includes: the type of building construction including but not limited to floors, walls, columns and roof assembly.
    - 13.4. *Exit access stairway* and *exit stairway* information that includes: number of *exit access stairways* and *exit stairways* in building; each *exit access stairway* and *exit stairway* designation and floors served; location where each *exit access stairway* and *exit stairway* discharges, *interior exit stairways* that are pressurized; *exit stairways* provided with emergency lighting; each *exit stairway* that allows reentry; *exit stairways* providing roof access; elevator information that includes: number of elevator banks, elevator bank designation, elevator car numbers and respective floors that they serve; location of elevator machine rooms, control rooms and control spaces; location of sky lobby; and location of freight elevator banks.
    - 13.5. Building services and system information that includes: location of mechanical rooms, location of building management system, location and capacity of all fuel oil tanks, location of emergency generator and location of natural gas service.

13.6. *Fire protection system* information that includes: location of standpipes, location of fire pump room, location of fire department connections, floors protected by automatic sprinklers and location of different types of *automatic sprinkler systems* installed including but not limited to dry, wet and pre-action.

13.7. Hazardous material information that includes: location and quantity of hazardous material.

14. Work table.
15. Generator supervision devices, manual start and transfer features.
16. Public address system, where specifically required by other sections of this code.
17. Elevator fire recall switch in accordance with ASME A17.1/CSA B44.
18. Elevator emergency or standby power selector switch(es) in accordance with ASME A17.1/CSA B44.

C The fire command center must contain all equipment necessary to enable the incident commander to monitor or control fire protection and other building service systems as listed in this section (also see commentary, Section 909.16). This room houses fire protection, smoke control and building system controls, as well as a workspace for emergency responders. The room also contains schematic plans and a work table so that responders have the basic layout and geometry of the building and can identify locations of utility controls, standby or emergency power systems, and where hazardous materials are stored or used. Providing concise information in a uniform format is essential to firefighters and emergency responders and improves their ability to utilize building systems to their advantage. This was confirmed during the National Institute of Science and Technology (NIST) investigations of the World Trade Center attacks on September 11, 2001. The Final Report on the collapse of the World Trade Center contained 30 key recommendations compiled by the NIST designed to address the building vulnerabilities learned in that tragedy. Three of those 30 recommendations embrace increasing situational awareness and emergency communications of first responders in large-scale emergencies. As a result of that investigation, this section includes an Item 13 that prescribes requirements for the Building Information Card (BIC).

The BIC is divided into multiple information areas and is intended to be formatted as a single form to provide a quick, concise source of information about the building. The code does not prescribe any particular format or layout for the BIC and does not have any limits on the level of information required to satisfy the requirements. It should be recognized that the BIC is intended to provide an easily understood and consistent tool to emergency responders who are taking control of systems in high-rise and smoke-protected assembly buildings. Jurisdictions should develop a policy to ensure that BICs are prepared in a standard, consistent format to avoid confusing the responders, and yet provide the minimum information required so they can correctly and efficiently utilize all of the building features. The number and types of features required by this section can create a large volume of data, thus reinforcing the need for an approved layout as required by Section 508.1.4.

508.1.7 Fire command center identification. The *fire command center* shall be identified by a permanent, easily visible sign stating "FIRE COMMAND CENTER" located on the door to the *fire command center*.

C Fire command centers are a critical feature for high-rise buildings, assemblies using mechanical smoke control for smoke-protected seating, and very large industrial buildings and storage warehouses. To make sure they are readily available for use during an emergency, they need to be easily located. Providing this signage is critical, especially for buildings that are spread out over a large area.

SECTION 509—FIRE PROTECTION AND UTILITY EQUIPMENT IDENTIFICATION AND ACCESS

509.1 Identification. Fire protection equipment shall be identified in an *approved* manner. Rooms containing controls for air-conditioning systems or *fire protection systems* shall be identified for the use of the fire department. *Approved* signs required to identify *fire protection system* equipment and equipment location shall be constructed of durable materials, permanently installed and readily visible.

C In an emergency, it is vitally important that the fire department and other emergency responders be able to quickly locate and access critical controls for fire protection systems. Obstructed or poorly marked equipment can cause delays in firefighting operations while firefighters locate other hose stations and stretch additional hose, for example. Valves and other controls are often located in rooms or other enclosures. Their location must be clearly identified with written or pictographic signs, which must be clearly visible and legible. Signs using the NFPA 170 symbols for fire protection equipment can provide standardized markings throughout a jurisdiction. White reflective symbols on a red reflective background are effective. For exterior signs, heavy-gage, sign-grade aluminum is recommended. Interior signs may be constructed of plastic, light-gage aluminum or other approved, durable, water-resistant material. As a general rule, fire protection piping, cabinets, enclosures, wiring, equipment and accessories are red or are identified by red or red/white markings. The manner of identification is subject to the approval of the fire code official.

509.1.1 Utility identification. Where required by the *fire code official*, gas shutoff valves, electric meters, service switches and other utility equipment shall be clearly and legibly marked to identify the unit or space that it serves. Identification shall be made in an *approved* manner, readily visible and shall be maintained.

C This section provides the fire code official with the authority to require utility identification for services serving multiple-unit or multiple-building properties, including facilities, campuses, strip malls, business parks, residential properties and similar locations where identification of utilities is considered essential to emergency responders. Note that this section does not prescribe any particular design requirements for utility identification markings. It should be recognized that the markings are intended to provide an easily understood and consistent tool to emergency responders who must secure utilities during emergency operations. Jurisdictions should develop a policy to ensure that utility identification markings are prepared in a standard, consistent format to avoid confusing the responders, and yet provide the minimum information required so they can correctly and efficiently utilize them. Note that the photovoltaic requirements in Section 1205.4 address signage to identify the rapid shutdown switches for these systems.

509.2 Equipment access. *Approved* access shall be provided and maintained for all *fire protection system* equipment to permit immediate safe operation and maintenance of such equipment. Storage, trash and other materials or objects shall not be placed or kept in such a manner that would prevent such equipment from ready access.

C This section requires immediate access to and working space around all fire suppression, protection and detection system devices and control elements necessary for fire department use. It further prohibits obstruction of such system equipment by materials or objects that may prevent such equipment from being immediately accessed by emergency responders.

SECTION 510—EMERGENCY RESPONDER COMMUNICATIONS ENHANCEMENT SYSTEMS

510.1 Emergency responder communications enhancement systems in new buildings. *Approved* in-building emergency responder communications enhancement system (ERCES) for emergency responders shall be provided in all new buildings. In-building ERCES within the building shall be based on the existing coverage levels of the public safety communications systems utilized by the jurisdiction, measured at the exterior of the building. The ERCES, where required, shall be of a type determined by the fire code official and the *frequency license holder(s)*. This section shall not require improvement of the existing public safety communications systems.

Exceptions:

1. Where *approved* by the building official and the *fire code official*, a wired communications system in accordance with Section 907.2.13.2 shall be permitted to be installed or maintained instead of an *approved* communications coverage system.
2. Where it is determined by the *fire code official* that the communications coverage system is not needed.
3. In facilities where emergency responder communications coverage is required and such systems, components or equipment required could have a negative impact on the normal operations of that facility, the *fire code official* shall have the authority to accept an automatically activated emergency responder communications coverage system.
4. One-story buildings not exceeding 12,000 square feet (1115 m^2) with no below-ground area(s).

C The provisions of Section 510 are concerned with the ability of emergency responders to communicate while inside buildings. More specifically, emergency responders need to communicate with others within the building and with emergency responders outside the building. The section does not specifically require an emergency responder communications enhancement system (ERCES); rather, it helps officials determine if one is needed. For example, such a system may be necessary based upon the assessment of signal strength needs within the building. Once the ERCES is required, this section provides the requirements associated with such systems. This is in keeping with the philosophy inherent in the I-Codes that, when a facility grows too large or complex for effective fire response, fire protection features must be provided within the building. While modeling and other techniques may provide a good prediction as to whether a building will interfere with communications, the reality is that it is unknown if a building will need to install communication enhancement systems until after the building is constructed. Determining factors may include construction type, shadows of other buildings, size of the structure, and use of building products such as low-emission glass. Though this section does not offer specific types of buildings that should be targeted, discussions with public safety radio professionals found that, based on current radio/communication technologies, these requirements should be applied in any building with one or more basements or below-grade building levels, underground buildings or buildings more than five stories in height.

Building construction features and materials can absorb or block the signal strength used to carry the signals inside or outside the building. Blockage or absorption of the signal can prevent a critical message from an emergency responder from being received and acknowledged. Depending on the incident, this loss of information can put other emergency responders at risk or may prevent an injured or disoriented emergency responder from communicating for assistance. The requirements apply to analog or digital radio systems and are applicable to all buildings.

This section requires that all buildings have approved communication coverage for emergency responders within the building. In other words, the signal strength prescribed in Section 510.4.1 must be achieved whether it is achieved with or without an ERCES.

Where testing using the existing public safety communications system finds that the signal strengths are not satisfactory, Exception 1 allows for the alternative installation of a wired communication system in accordance with

Section 907.2.13.2, which requires that such a system be designed in accordance with NFPA 72. When applying this exception, the concurrent approval of fire and building code officials is required.

Where testing using the existing public safety communications system finds that the signal strengths are satisfactory, Exception 2 allows the fire code official to waive the requirements when it is determined that an in-building, emergency responder communications enhancement system is not needed. This exception does not give any criteria as to buildings that can be exempted.

Exception 3 provides a means for the fire code official to allow the installation of a manual or automatic switch that turns on the in building ERCES when it is needed. Exception 3 also recognizes that operating ERCES in certain buildings with electronic equipment sensitive to radio frequency (RF) energy can cause damage to the equipment, or worse, impact the operations of a local or regional computer network. One such occupancy is a telephone CO, which is where landline and cellular telephone signals are received and dispatched to the recipient caller. It is not uncommon for a telephone CO to be capable of receiving and processing more than 250,000 telephone calls within a 1-minute period. Telephone COs serve an important public safety function because they process emergency or information calls routed via 911 to a jurisdiction's public safety answering point.

Harmful interference (noise) can be detrimental to a community's public safety communications network. When planning for communications coverage, it is vital that solutions only be installed where truly needed. Simply putting a signal booster in small buildings with relatively short travel distances for first responders to reach the outside of buildings must be factored into the equation. Many communities have instituted minimum thresholds that must be met before requiring ERCES. Based upon these concerns, a threshold of one-story buildings not exceeding 12,000 square feet (1115 m^2) and without underground areas is provided in Item 4. The trigger of 12,000 square feet is utilized in other areas of this code, such as within Section 903.3 and Table 3206.2 for high-piled combustible storage.

510.2 Emergency responder communications enhancement system in existing buildings. Existing buildings shall be provided with *approved* in-building emergency responder communications enhancement system for emergency responders as required in Chapter 11.

C See the commentary to Section 1103.2.

510.3 Permits. Permits for in-building emergency responder communications enhancement systems shall be in accordance with Sections 510.3.1 and 510.3.2.

C This section simply introduces the necessary construction and operational permits associated with the installation and ongoing use of ERCES.

510.3.1 Permit required. A construction permit for the installation of or modification to in-building emergency responder communications enhancement systems and related equipment is required as specified in Section 105.6.5. Maintenance performed in accordance with this code is not considered a modification and does not require a permit.

C A construction permit must be obtained in accordance with Section 105.6.5 prior to the installation of the in-building, two-way emergency responder communication systems and for any modification or alteration to the system to ensure that the work is done correctly and any parts replacement will be compatible with the existing system components. Note that normal maintenance required for the system would not require a permit.

510.3.2 Operational permit. Where required by the fire code official, an operational permit shall be issued for the operation of an in-building emergency responder communications enhancement system.

C Equally important as the construction permit is an operational permit to address the ongoing use of ERCES with annual renewal. Renewable permits and written authorization by the frequency license holder are two components that need to be addressed for the life of the ERCES in order to comply with the retransmitting of a licensed frequency. When written authorization from the frequency license holder(s) is required by the authority granting the license, it shall be obtained before activating the ERCES. A renewable permit provides a method to maintain the required authorization.

510.4 Technical requirements. Equipment required to provide in-building emergency responder communications enhancement shall be *listed* in accordance with UL 2524. Systems, components and equipment required to provide the in-building emergency responder communications enhancement system shall comply with Sections 510.4.1 through 510.4.2.9.

C This section requires listed equipment to be installed to address fire safety and shock safety of ERCES. This also provides further compliance with the performance requirements specified in this section and NFPA 1225. Additionally, this section introduces Sections 510.4.1 through 510.4.2.9, which are the technical requirements for ERCES.

510.4.1 Emergency responder communications enhancement system signal strength. The building shall be considered to have an acceptable in-building emergency responder communications enhancement system where signal strength measurements in 95 percent of all areas and 99 percent of areas designated as *critical areas* by the *fire code official* on each floor of the building meet the signal strength requirements in Sections 510.4.1.1 through 510.4.1.3.

C This section introduces the minimum acceptable signal criteria that must be achieved and maintained throughout 95 percent of all areas on each floor of a building, as indicated in Sections 510.4.1.1 through 510.4.1.3. In addition, this section reflects the requirements of NFPA 1225 to provide a stronger signal in critical areas. Covering critical areas of a building is vital to the operations of public safety responders. Critical areas are defined in Section 202. Such areas are designated as needing the highest level of emergency responder communication coverage. Types of areas considered critical include exit stairways, exit passageways, elevator lobbies, equipment rooms and fire command centers. In some cases, there may be additional areas that are considered critical for communication.

510.4.1.1 Minimum signal strength into the building. The minimum *downlink* signal strength shall be sufficient to provide usable voice communications throughout the coverage area as specified by the *fire code official*. The *downlink* signal level shall be sufficient to provide not less than a Delivered Audio Quality (DAQ) of 3.0 throughout the coverage area using either narrowband analog, digital or wideband LTE signals or an equivalent bit error rate (BER), or signal-to-interference-plus-noise ratio (SINR) applicable to the technology for either analog or digital signals.

C The focus of this section is the signal received within the building. Section 510.4.1.2 addresses the minimum downlink signal to a radio on the outside of the building. This section requires that the downlink be sufficient to provide a DAQ of 3.0. Delivered audio quality (DAQ) refers to a range of usable voice parameters and is useful regardless of the modulation or system technology utilized. This allows a measure of how the signal will sound to the end user, which is critical to emergency operations. This is the same criteria provided for minimum outbound signal strength in Section 510.4.1.2.

510.4.1.2 Minimum signal strength out of the building. The minimum *uplink* signal strength shall be sufficient to provide usable voice communications throughout the coverage area as specified by the *fire code official*. The *uplink* signal level shall be sufficient to provide not less than a delivered audio quality (DAQ) of 3.0 using either narrowband analog, digital or wideband LTE digital signals or an equivalent bit error rate (BER), or an equivalent SINR applicable to the technology for either analog or digital signals.

C The criteria for the uplink signal are similar to the downlink signal. See the commentary to Section 510.4.1.1.

510.4.1.3 System performance. Signal strength shall be sufficient to meet the requirements of the applications being utilized by public safety for emergency operations through the coverage area as specified by the *fire code official* in Section 510.4.2.2.

C This section addresses data network performance for other emergency responder signals such as Long Term Evolution (LTE), which is part of the nationwide public safety responder network commonly known as FirstNet.

510.4.2 System design. The in-building emergency responder communications enhancement system shall be designed in accordance with Sections 510.4.2.1 through 510.4.2.9 and NFPA 1225.

C This section introduces Sections 510.4.2.1 through 510.4.2.9 as the system design criteria for in-building, emergency responder communications enhancement systems. NFPA 1225 is the standard that addresses installation, maintenance and use of emergency services communication systems.

510.4.2.1 Amplification systems and components. Buildings and structures that cannot support the required level of in-building emergency responder communications enhancement system shall be equipped with systems and components to enhance the radio signals and achieve the required level of in-building emergency responder communications enhancement system specified in Sections 510.4.1 through 510.4.1.3. In-building emergency responder communications enhancement systems utilizing radio-frequency-emitting devices and cabling shall be *approved* by the *fire code official*. Prior to installation, all RF-emitting devices shall have the certification of the radio licensing authority and be suitable for public safety use.

C In many cases, buildings cannot provide the necessary inbound (downlink) and outbound (uplink) performance levels without providing amplification within the building. This section stresses that additional amplification must be provided to achieve the necessary performance. There are several methods to provide the amplification needed by emergency responders in structures identified as needing ERCES. These enhancements need to be both approved by the fire code official and also through certification through the radio licensing authority. In the case of the United States, this would be the Federal Communications Commission (FCC). The wording is more performance based than previous editions to state more clearly that a variety of different methods can be used.

510.4.2.2 Technical criteria. The *fire code official* shall maintain a document providing the specific technical information and requirements for the in-building emergency responder communications enhancement system. This document shall contain, but not be limited to, the various frequencies required, the location of radio sites, the effective radiated power of radio sites, the maximum propagation delay in microseconds, the applications being used and other supporting technical information necessary for system design.

C In order to effectively install such systems, the fire code official must provide basic information such as the frequency range to be supported and maximum propagation delay. For example, a fire department may provide a requirement such as "The frequency ranges that must be supported shall be 806 MHz to 824 MHz and 851 MHz to 869 MHz." The fire code official will likely need to provide a reference to the local communications center or provide details on radio sites in the jurisdiction.

510.4.2.3 Standby power. In-building emergency responder communications enhancement systems shall be provided with dedicated standby batteries or provided with 2-hour standby batteries and connected to the facility generator power system in accordance with Section 1203. The standby power supply shall be capable of operating the in-building emergency responder communications enhancement system at 100 percent system capacity for a duration of not less than 12 hours.

C This section requires secondary power to operate the equipment in cases where the primary building power must be shut down or is lost. There are two options provided: dedicated standby batteries or through a traditional generator-type source with 2-hour standby batteries. The standby power supply, regardless of how it is provided, must be capable of operating the ERCES for a minimum 12-hour duration at 100 percent system capacity. The 12-hour value was selected to ensure the reliability of the signal boosters during long-term emergency operations such as response to natural disasters where utility-supplied electrical power is disabled. See Section 1203 for secondary power system specifics and standards. This is consistent with NFPA 1225.

510.4.2.4 Signal booster requirements. If used, signal boosters shall meet the following requirements:

1. All signal booster components shall be contained in a NEMA Type 4 cabinet.
2. Battery systems used for the emergency power source shall be contained in a NEMA 3R or higher-rated cabinet.
3. Equipment shall have FCC or other radio licensing authority certification and be suitable for public safety use prior to installation.
4. Where a donor antenna exists, isolation shall be maintained between the donor antenna and all inside antennas to not less than 20dB greater than the system gain under all operating conditions.
5. Active RF-emitting devices used for in-building emergency responder communications enhancement systems shall have built-in oscillation detection and control circuitry to reduce gain and maintain operation. When a signal booster detects oscillation, a supervisory signal shall be transmitted. In the event of uncorrectable oscillation, the system shall be permitted to shut down.
6. The installation of amplification systems or systems that operate on or provide the means to cause interference on any in-building emergency responder communications enhancement network shall be coordinated and *approved* by the *fire code official* and the *frequency license holder(s)*.

C If a building is equipped with a signal booster, this section requires that the components of the signal booster be located in a NEMA Type 4 cabinet. A NEMA Type 4 cabinet is designed to protect personnel having incidental contact with the equipment and to protect the equipment from falling dirt, rain, snow, windblown dust, and both splashing and hose-directed water. The standby power source is to be located in a NEMA Type 3R cabinet, which provides similar protection but with the ventilation necessary for most batteries. Type 3R does not require protection from splashing water and hose-directed water but is felt to provide an appropriate level of protection. The system is also required to be certified by the FCC or other radio licensing authority. This document may be used outside the United States where the FCC is not applicable. Items 4, 5 and 6 provide requirements that limit the opportunity for interference and/or noise created by inadequate system components and their location and placement.

510.4.2.5 System monitoring. The in-building emergency responder communications enhancement system shall be monitored by a *listed fire alarm control unit*, or where *approved* by the *fire code official*, shall sound an audible signal at a constantly attended on-site location. Automatic supervisory signals shall include the following:

1. Loss of normal AC power supply.
2. System battery charger(s) failure.
3. Signal source malfunction.
4. Failure of active RF-emitting device(s).
5. Low-battery capacity at 70 percent of the 12-hour operating capacity has been depleted.
6. Failure of critical system components.
7. The communications link between the *fire alarm system* and the in-building emergency responder communications enhancement system.
8. Oscillation of active RF-emitting device(s).

C This section lists the specific aspects of the system that are required to be monitored. These requirements are those necessary to maintain integrity of the ERCES.

510.4.2.5.1 Single supervisory input. Where *approved*, a single supervisory input to the *fire alarm system* to monitor all system supervisory signals shall be permitted.

C This section allows the fire alarm system to receive a single supervisory signal—rather than the eight distinct signals required by Section 510.4.2.5—where approved by the fire code official.

510.4.2.6 Additional frequencies and change of frequencies. The in-building emergency responder communications enhancement system shall be capable of modification or expansion in the event *frequency* changes are required by the FCC or other frequency licensing authorities, or additional frequencies are made available by the FCC or other frequency licensing authorities.

C Because of potential changes in public safety frequency bands, the code requires the capability for changing the frequency in the future. Note that since this code can be applied outside the United States, reference is made to the FCC or other frequency licensing authority.

510.4.2.7 Design documents. The *fire code official* shall have the authority to require "as-built" design documents and specifications for in-building emergency responder communications enhancement systems. The documents shall be in a format acceptable to the *fire code official*.

C This information is readily available as documentation typically exists electronically where systems are designed and installed. Many jurisdictions utilize this type of information within their electronic records management systems and computer-aided dispatch systems. This section is consistent with other construction document requirements in Chapter 9.

510.4.2.8 Near-far effect. Where a signal booster is required by the RF system designer, the dynamic range of the in-building emergency responder communications enhancement system shall be designed to minimize the effects of strong signal automatic gain control on weak signal *uplink* performance.

C The near-far effect occurs where too few indoor antennas are used to enhance coverage inside the building, creating an excessively wide dynamic range of operation. A portable device in close proximity to an indoor antenna, when keyed, can cause the talk-out amplifier's automatic gain control/automatic level control (AGC/ALC) circuit to reduce the gain, leaving other portables farther away at risk of not hitting the repeater site due to insufficient gain. If the near-far effect occurs, some public safety communications equipment will not function as required by the code and will leave responders at risk.

510.4.2.9 Noise interference. Where a signal booster is used, signal booster type(s) and the *uplink* signal and noise levels shall be coordinated with and *approved* by all *frequency license holder(s)* that may be adversely impacted by any transmitted noise resulting from the in-building emergency responder communications enhancement system. Systems shall be in compliance with all *frequency licensing authority* requirements.

C When solving the communications coverage issues within a building, it is vital to have a full understanding of the actual public safety communication systems that are being utilized within the coverage area. The frequency license holder of those radio frequencies (RF) must be involved in determining which solution, if any, can be utilized to enhance RF without creating harmful interference. Many have a false belief that these systems are simple plug-and-play solutions. When signal boosters are selected as the solution to solve the in-building communications problem, they can and do create harmful interference (otherwise known as noise) on the public-safety macro communications system, rendering it inoperable for the entire community and all emergency responders.

510.5 Installation requirements. The installation of the in-building emergency responder communications enhancement system shall be in accordance with NFPA 1225 and Sections 510.5.2 through 510.5.5.

C This section simply introduces the installation requirements for such systems. In addition, it also notes that such systems must also comply with NFPA 1225. NFPA 1225 addresses installation, maintenance and use of emergency services communication systems, including ERCES.

510.5.1 Mounting of the donor antenna(s). To maintain proper alignment with the system designed donor site, donor antennas shall be permanently affixed on the building or where *approved* , mounted on a movable sled with a clearly visible sign stating "MOVEMENT OR REPOSITIONING OF THIS ANTENNA IS PROHIBITED WITHOUT APPROVAL FROM THE FIRE CODE OFFICIAL*."* The antenna installation shall be in accordance with the applicable requirements in the *International Building Code* for weather protection of the building envelope.

C The location and positioning of the donor antenna is critical to the successful operation of ERCES. There are two basic methods of mounting donor antennas, either fixed or on an antenna sled held down by weights. Mounting the donor antenna in a fixed position limits the amount of movement that may occur. Although antenna sleds are common practice for mounting various types of antennas on a roof, those utilized for ERCES should be approved by the fire code official. Adding signage alerts others such as roofers and HVAC repair personnel to secure approval prior to moving or repositioning the antenna to avoid potentially rendering the ERCES inoperable.

510.5.2 Approval prior to installation. Communications enhancement systems capable of operating on frequencies licensed to any public safety *agency* by the FCC or other *frequency licensing authority* shall not be installed without prior coordination and approval of the *fire code official* and *frequency license holder.*

C The installation of ERCES is similar to other building features where review of the plans and installation is required and covered under a permit. Part of the approval process for amplification systems operating on frequencies licensed to a public safety agency requires approval of the local fire code official.

Involving the frequency license holder throughout the process allows those legally responsible for operating the public safety macro communications system to determine the overall impact the solution will have on their system. They can then make recommendations that provide a functional solution. In the United States, it is a federal requirement that the frequency license holder provides written consent to activate a signal booster on the communications system. An improperly designed and installed signal booster system results in harmful noise/interference. Section 510.4.2.9, which is related to noise/interference, provides direction to ensure the proper solution has been selected. The frequency license holder must also have knowledge of the location of other signal boosters. Where multiple signal boosters are placed too close to each other, harmful interference occurs.

510.5.2.1 Active RF-emitting devices. Active RF-emitting devices shall meet the following requirements in addition to any other requirements determined by the *fire code official* or the *frequency license holder(s)*:

1. Active RF-emitting devices that have a transmitted power output sufficient to require certification of the *frequency licensing authority* shall have the certification of the radio *frequency licensing authority* prior to installation.
2. All active RF-emitting devices shall be simultaneously compatible for their intended use, as required by the *frequency licensing authority*, the *frequency license holder(s)* and the *fire code official,* at the time of installation.
3. Written authorization shall be obtained from the *frequency license holder(s)* prior to the initial activation of any RF-emitting devices required to be certified by the *frequency licensing authority*.

C This section clarifies that documented permission must be obtained in order to retransmit or broadcast on a licensed frequency. Failure to do so is in violation of laws stipulated by the Federal Communications Commission (FCC). It is imperative that there be communications between the frequency license holder and the fire code official prior to installing and operating any enhancement system. Additionally, it is imperative that active RF-emitting devices, where used, are certified by the frequency licensing authority.

510.5.3 Minimum qualifications of personnel. The minimum qualifications of the system designer and lead installation personnel shall include both of the following:

1. A valid FCC-issued general radio operators license.
2. Certification of in-building system training issued by an *approved* organization or *approved* school, or a certificate issued by the manufacturer of the equipment being installed.

These qualifications shall not be required where demonstration of adequate skills and experience satisfactory to the *fire code official* is provided.

C This section ensures that a qualified person designs and installs the ERCES. The qualification may be from a school program in communications or a manufacturer's training program. This section also allows the jurisdiction to accept a person or business upon adequate demonstration of skills and experience to the fire code official.

510.5.4 Acceptance test procedure. Where an in-building emergency responder communications enhancement system is required, and upon completion of installation, the building *owner* shall have the radio system tested to verify that two-way coverage on each floor of the building is not less than 95 percent. The test procedure shall be conducted as follows or by a method *approved* by the *fire code official*:

1. Each floor of the building shall be divided into a grid of 20 approximately equal test areas.
2. The test shall be conducted using a calibrated portable radio of the latest brand and model used by the *agency* talking through the *agency*'s radio communications system or equipment *approved* by the *fire code official*.
3. Failure of more than one test area shall result in failure of the test.
4. In the event that two of the test areas fail the test, in order to be more statistically accurate, the floor shall be permitted to be divided into 40 equal test areas. Failure of not more than two nonadjacent test areas shall not result in failure of the test. If the system fails the 40-area test, the system shall be altered to meet the 95 percent coverage requirement.
5. A test location approximately in the center of each test area shall be selected for the test, with the radio enabled to verify two-way communications to and from the outside of the building through the public *agency*'s radio communications system. Once the test location has been selected, that location shall represent the entire test area. Failure in the selected test location shall be considered to be a failure of that test area. Additional test locations shall not be permitted.
6. The gain values of all amplifiers shall be measured and the test measurement results shall be kept on file with the building *owner* so that the measurements can be verified during annual tests. In the event that the measurement results become lost, the building *owner* shall be required to rerun the acceptance test to reestablish the gain values.
7. As part of the installation, a spectrum analyzer or other suitable test equipment shall be utilized to ensure spurious oscillations are not being generated by the subject signal booster. This test shall be conducted at the time of installation and at subsequent annual inspections.
8. Systems shall be tested using two portable radios simultaneously conducting subjective voice quality checks. One portable radio shall be positioned not greater than 10 feet (3048 mm) from the indoor antenna. The second portable radio shall be positioned at a distance that represents the farthest distance from any indoor antenna. With both portable radios simultaneously keyed up on different frequencies within the same band, subjective audio testing shall be conducted and comply with DAQ levels as specified in Sections 510.4.1.1 and 510.4.1.2.

C This section provides a testing procedure to be followed upon completion of system installation and prior to acceptance of the system by the jurisdiction. The building owner or agent is responsible for ensuring that the ERCES is functioning properly prior to the building being occupied. Note that if there are problems as a result of the testing, the test area is to be altered to provide more specific data on the strength of the signal-boosting capability of the system. Testing results are to be kept on file by the building owner for annual testing verification. If the testing results are lost, a retest of the building shall be required to determine compliance levels. The testing identified in Item 8 will ensure that there is consistent, objective data for the fire code official to use to ensure quality and conformance and to maintain consistency with the DAQ requirements in Section 510.4.

510.5.5 FCC compliance. The in-building emergency responder communications enhancement system installation and components shall comply with all applicable federal regulations including, but not limited to, FCC 47 CFR Part 90.219.

C As with all radio systems, the ERCES and its components must comply with all applicable FCC regulations.

510.6 Maintenance. The in-building emergency responder communications enhancement system shall be maintained operational at all times in accordance with Sections 510.6.1 through 510.6.4.

C This section simply introduces the maintenance requirements for the ERCES for continued successful operation of the system.

510.6.1 Testing and proof of compliance. The *owner* of the building or *owner*'s authorized agent shall have the in-building emergency responder communications enhancement system inspected and tested annually or where structural changes occur, including additions or remodels that could materially change the original field performance tests. Testing shall consist of the following:

1. In-building coverage test as described in Section 510.5.4.
2. Signal boosters shall be tested to verify that the gain is the same as it was upon initial installation and acceptance or set to optimize the performance of the system.
3. Backup batteries and power supplies shall be tested under load of a period of 1 hour to verify that they will properly operate during an actual power outage. If within the 1-hour test period the battery exhibits symptoms of failure, the test shall be extended for additional 1-hour periods until the integrity of the battery can be determined.
4. All active components shall be checked to verify operation within the manufacturer's specifications.

At the conclusion of the testing, a report, which shall verify compliance with Section 510.5.4, shall be submitted to the *fire code official.*

C This section is focused on key aspects that must be tested annually to provide for a functioning system on an ongoing basis. Signal boosters are tested to verify that the gain is equal to that produced during the initial acceptance. Testing must also include functional tests of the secondary power supply and an inspection of any other components connected to the in-building amplification system. The inspection report must be submitted to the fire code official and requires documentation that the amplification system complies with the requirements in Section 510.5.4 This section also authorizes the fire code official to require additional tests where structural changes or modifications occur that could materially change the performance of the signal boosting system. This section also emphasizes that the responsibility to undertake this testing is that of the owner.

510.6.2 Additional frequencies. The building *owner* shall modify or expand the in-building emergency responder communications enhancement system at their expense in the event *frequency* changes are required by the FCC or other radio licensing authority, or additional frequencies are made available by the FCC or other radio licensing authority. Prior approval of an in-building emergency responder communications enhancement system on previous frequencies does not exempt this section.

C Where additional frequencies are needed to change or expand the coverage of the ERCES, the building owner is the person responsible for ensuring that the work is performed. Previous approvals of the system do not apply when changes are needed. Changes to the system are not exempt because of prior approval. If the change is needed, the work must be done.

510.6.3 Nonpublic safety system. Where other nonpublic safety amplification systems installed in buildings reduce the performance or cause interference with the in-building emergency responder communications enhancement system, the nonpublic safety amplification system shall be corrected or removed.

C With the public's reliance on cellular devices as a primary method of communications, many buildings are being equipped with cellular enhancement systems that provide improved coverage. If not properly designed, installed and maintained, these nonpublic safety systems may cause interference and performance issues on the public safety communication enhancement system. This section provides the necessary tool for the fire code official to address interference with a required public safety system. Requiring correction or removal of the nonpublic safety system where it is causing interference or performance issues is vital to the public safety responders in the event of an incident.

510.6.4 Field testing. *Agency* personnel shall have the right to enter onto the property at any reasonable time to conduct field testing to verify the required level of radio coverage.

C Like all systems, ERCES need to be tested to verify their continued sufficiency. Whether the test is an annual review of the system or an in-service familiarization test by an engine, truck or squad company, this section provides the ability of the personnel to enter the building at reasonable hours to test and operate the system. As with any inspection, the right of entry for testing the system is limited by constitutional constraints. See the commentary to Section 104.4 for further discussion on the right of entry.

Bibliography

The following resource materials were used in the preparation of the commentary for this chapter of the code:

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